HPV vaccination simulation model

Cost-benefit analysis of extending vaccination age cap within VHA
Updated slides based on new transition model.

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Agent-based models

- Step 1: Simulate heterogenous population of agents
  - **Agents**: with features.
  - **Men in VA**: tobacco use + vaccine hesitancy + age
- Step 2: Model actions and interactions with environment
  - **Environment and Interactions**: MSM + vaccine hesitancy.
- Step 3: Flexible tracking of trajectory over time.
  - **Behavior changes**: vaccine uptake + smoking + MSM.

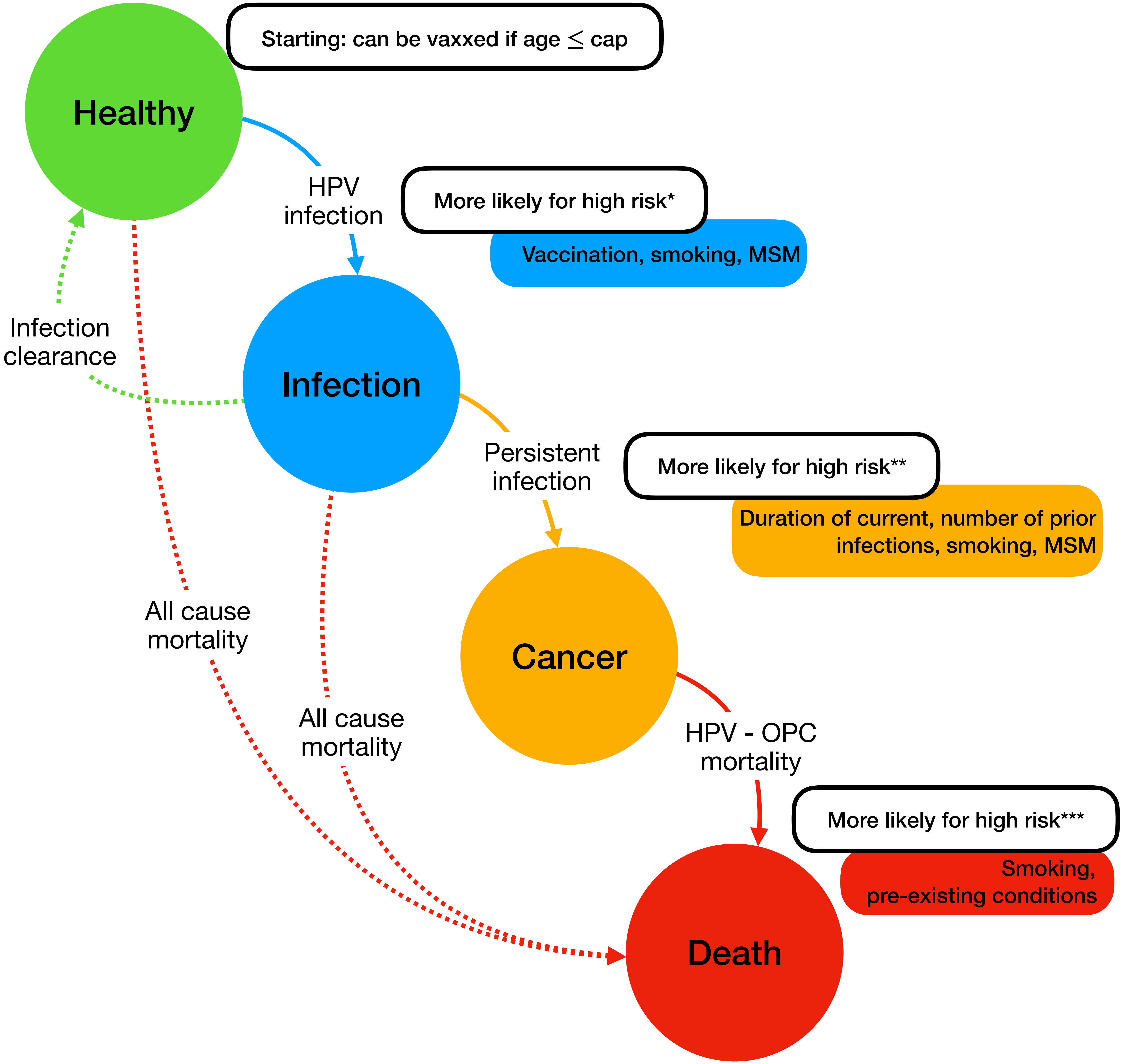
Designing an ABM

Attributes of agent

1. $20 \leq$ **Age** ≤ 85**:** will simulate according to VA.
2. **Vaccination status:** Y/N: will simulate according to VA.
3. **HPV status:** Positive/Negative: will need input.
4. **Duration of infection (in years):** ≥ 0
5. **Number of previous infections:** ≥ 0
6. **Covariates:** can change over time
    1. *Vaccine hesitancy*
    2. *Smoking history*
    3. *MSM behavior**: # of sexual partners (?)

* have not included in model for now.

Primary pathway of interest: solid lines



Designing an ABM

Generate starting population and run simulation for 60 cycles.
For a fixed cycle: iterate through EACH living person in population.

Iteration and updating rule:

1. If healthy:
    1. If **unvaccinated:**
        1. Simulate vaccination (under age cap), go to next year
        2. Simulate infection (**high** risk), go to next year
        3. Simulate death (all-cause mortality), leave population
    2. If vaccinated:
        1. Simulate infection (low risk), go to next year
        2. Simulate death (all-cause mortality), leave population
2. If infected:
    1. Simulate return to healthy: infection clearance, go to next year
    2. Simulate infection, go to next year
    3. Simulate cancer (high risk), go to next year
    4. Simulate death (all-cause mortality), leave population
3. If cancer:
    1. Cancer persists, go to next year.
    2. Simulate death (high risk): OPC mortality, leave population